Assignment 2

Section C

1. A college stores student and course data. Identify tables and keys.

Tables

1. Student

Attribute	**Key Type**
Student_ID	Primary Key
Name	
Email	
Phone	

2. Course

Attribute	**Key Type**
Course_ID	Primary Key
Course_Name	
Credits	

3. Enrollment

Attribute	**Key Type**
Enrollment_ID	Primary Key
Student_ID	Foreign Key
Course_ID	Foreign Key

Keys Explanation

Primary Keys

- Student_ID uniquely identifies each student
- Course_ID uniquely identifies each course
- Enrollment_ID uniquely identifies each enrollment record

Foreign Keys

- Student_ID in Enrollment references Student(Student_ID)
- Course_ID in Enrollment references Course(Course_ID)

Relationship

- Student and Course have a many-to-many (M:N) relationship
- This is resolved using the Enrollment table

2. A bank stores customer and account data. Suggest constraints.

Entities

1. Customer
   - Customer_ID (Primary Key)
   - Name
   - Address
   - Contact

2. Account
   - Account_ID (Primary Key)
   - Account_Type
   - Balance
   - Customer_ID (Foreign Key)

Constraints

Primary Key Constraint

- Customer_ID and Account_ID must be unique and not NULL

Foreign Key Constraint

- Customer_ID in Account must reference Customer(Customer_ID)
- Ensures referential integrity

NOT NULL Constraint

- Important fields like Name, Account_Type, and Balance cannot be NULL

UNIQUE Constraint

- Contact number of customer can be unique

CHECK Constraint

- Balance must be greater than or equal to zero
- Account_Type must be from predefined values (e.g., Savings, Current)

DEFAULT Constraint

- Default balance can be set to 0

3. An online store stores product prices. Which constraint will be used?

Constraint Used: CHECK

Explanation

The CHECK constraint is used to ensure that the values in a column satisfy a specific condition. In this case, it is used to ensure that product prices are valid.

Example

- Price must be greater than zero
- Prevents invalid entries such as negative or zero price

SQL

```
Price DECIMAL(10,2) CHECK (Price > 0)
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